EXPLORATORY DATA ANALYSIS (EDA)

Used Car Dataset Analysis

CarDekho used-car dataset



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Individual project | Unlox Student Project Assignment

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on a used-car dataset obtained from Kaggle. The analysis aims to understand the dataset, examine its quality, identify patterns and outliers, and explore the relationships between different features affecting used-car prices.

The main objectives are:

- To understand the structure and characteristics of the dataset.
- To identify and handle missing values, duplicate records, and outliers.
- To analyse the distribution of car prices and identify the most common car brands and fuel types.
- To examine the relationship between vehicle age, kilometres driven, mileage, engine power, and selling price.
- To compare the average selling prices of automatic and manual cars.
- To study correlations between numerical variables using statistical methods and visualisations.

Dataset Overview

The dataset used in this project is the CarDekho Used-Car Dataset, downloaded from Kaggle as a publicly available dataset. It contains 15,411 records and 14 columns, with each row representing a used-car listing.

The dataset includes information about car brands, models, vehicle age, kilometres driven, fuel type, transmission type, mileage, engine capacity, maximum power, seating capacity, and selling price.

Features of the Dataset

The dataset contains two main types of variables:

1. Numerical Variables:

These include vehicle age, kilometres driven, mileage, engine capacity, maximum power, number of seats, and selling price. They help us perform statistical analysis and understand relationships between different features.

2. Categorical Variables:

These include car name, brand, model, seller type, fuel type, and transmission type. They help us analyse the distribution of cars across different categories.

- Dataset Details
- Dataset Name: CarDekho Used-Car Dataset
- Source: Kaggle
- Number of Records: 15,411
- Number of Columns: 14
- Target Variable of Interest: Selling Price
- Tools Used: Python, Pandas, Matplotlib, Seaborn, and Jupyter Notebook
- Purpose of Dataset Analysis

The dataset is used to investigate how different vehicle characteristics are associated with selling prices. Through data cleaning, statistical summaries, correlation analysis, and visualisations, the project aims to identify meaningful patterns and present the findings clearly.

Column	Description	Data type	Kind
Unnamed: 0	Row index from the source file (not an analytical feature)	int64	Numerical
car_name	Brand and model name combined	object	Categorical
brand	Manufacturer of the car	object	Categorical
model	Car model	object	Categorical
vehicle_age	Age of the car in years	int64	Numerical
km_driven	Distance driven in kilometres	int64	Numerical
seller_type	Type of seller (e.g. Individual, Dealer)	object	Categorical
fuel_type	Petrol, Diesel, CNG, LPG or Electric	object	Categorical
transmission_type	Manual or Automatic	object	Categorical
mileage	Fuel efficiency (assumed km per litre)	float64	Numerical
engine	Engine displacement (assumed cc)	int64	Numerical
max_power	Maximum engine power (assumed bhp)	float64	Numerical
seats	Number of seats	int64	Numerical
selling_price	Listed selling price (assumed INR); the target variable of interest	int64	Numerical

Sample Records

car_name	Age	km_driven	Fuel	Trans.	Mileage	Engine	Max power	Seats	Price
Maruti Alto	9	120,000	Petrol	Manual	19.70	796	46.30	5	120,000
Hyundai Grand	5	20,000	Petrol	Manual	18.90	1197	82.00	5	550,000
Hyundai i20	11	60,000	Petrol	Manual	17.00	1197	80.00	5	215,000
Maruti Alto	9	37,000	Petrol	Manual	20.92	998	67.10	5	226,000
Ford Ecosport	6	30,000	Diesel	Manual	22.77	1498	98.59	5	570,000

Data Quality and Cleaning

The following checks were run before any analysis. Each decision and its reason is recorded below.

Check	Result	Decision and reason
Missing values	0 missing values in all 14 columns	No imputation or row removal was needed.
Duplicate rows	0 duplicate rows	Nothing removed.
Data types	All columns already have suitable types (integers and floats for numeric data, text for categories)	No type conversion was required.
Inconsistent labels	Value counts of fuel_type, transmission_type and brand showed clean, consistent labels	No relabelling was needed for these columns.
Identifier column	'Unnamed: 0' is a row index from the source file	Not meaningful for analysis; ignored when interpreting results (it shows about zero correlation with every variable).
Outliers in selling_price	IQR rule (Q1 - 1.5 x IQR, Q3 + 1.5 x IQR) flagged 1,386 listings, about 9.0% of the data. The highest price before cleaning is about 3.95 crore (3.95 x 107).	Rows outside the bounds were removed to stop a few very expensive cars from distorting averages and charts. Trade-off: premium and luxury cars are excluded from the later charts.

Dataset size before and after cleaning

Stage	Rows	Columns
Before cleaning (raw data)	15,411	14
After outlier removal on selling_price	14,025	14
Rows removed	1,386 (about 9.0%)	0

Descriptive Analysis

Numerical variables

Statistics below are calculated on the full dataset (15,411 rows) before outlier removal.

Variable	Mean	Median	Std. deviation
vehicle_age (years)	6.04	6	3.01
km_driven	55,616	50,000	51,619
mileage	19.70	19.67	4.17
engine	1,486	1,248	521
max_power	100.59	88.50	42.97
seats	5.33	5	0.81
selling_price	774,971	556,000	894,128

Observations: The mean selling price (about 7.75 lakh) is well above the median (5.56 lakh) and its standard deviation is larger than the mean, so price is strongly right-skewed. km_driven and engine show the same pattern (mean above median). Vehicle age, mileage, max_power and seats are close to symmetric, with the mean near the median.

Categorical variables

Fuel type	*Count*	*Share*	*Transmission*	*Count*	*Share*
Petrol	7,643	49.6%	Manual	12,225	79.3%
Diesel	7,419	48.1%	Automatic	3,186	20.7%
CNG	301	2.0%			
LPG	44	0.3%			
Electric	4	0.03%			

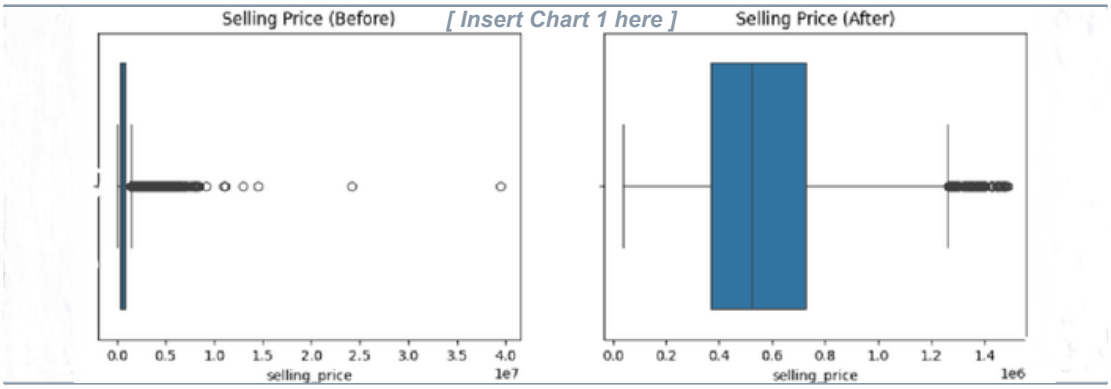
Top 10 brands	*Maruti*	*Hyundai*	*Honda*	*Mahindra*	*Toyota*	*Ford*	*VW*	*Renault*	*BMW*	*Tata*
Listings	4,992	2,982	1,485	1,011	793	790	620	536	439	430

Observations: Petrol and diesel together make up about 97.7% of listings. Manual cars are roughly four times as common as automatics. Maruti alone accounts for about 32% of listings, and Maruti, Hyundai and Honda together for about 61%.

Visual EDA

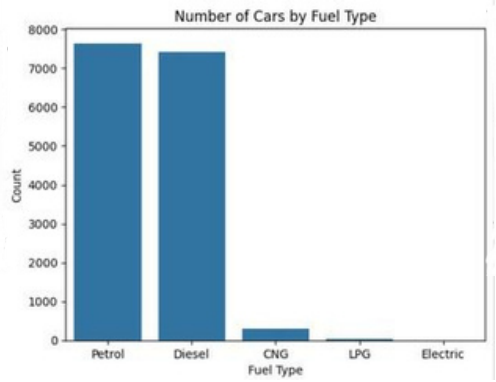
Six charts were produced, one each of these types: box plot, count plot, histogram, scatter plot, bar plot and heatmap. Chart 1 compares the raw and cleaned data, Chart 2 uses the raw data (15,411 rows), and Charts 3 to 6 use the cleaned data (14,025 rows). Prices are in the dataset's currency (assumed INR).

Chart 1: Selling Price Before and After Outlier Removal



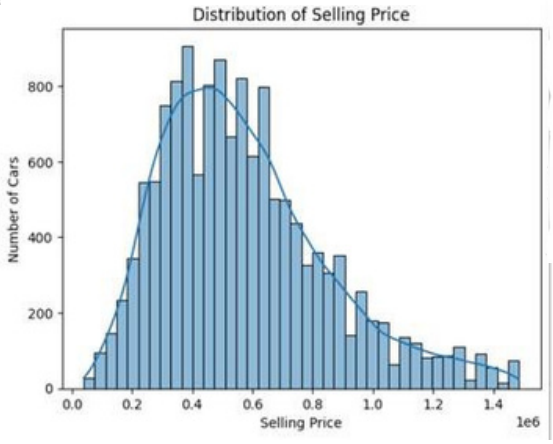
Observation: Before cleaning, the box is squeezed against zero because a long tail of expensive cars stretches the axis out to about 4 crore. After removing outliers the box and whiskers become readable: the middle 50% of cars sit roughly between 3.7 and 7.3 lakh. A few points still appear above the upper whisker because the quartiles were recalculated on the smaller dataset.

Chart 2: Number of Cars by Fuel Type (Count Plot)



Observation: Petrol and diesel are almost equally common (about 7,600 and 7,400 cars). CNG, LPG and Electric bars are barely visible, with only 301, 44 and 4 listings respectively, so conclusions about these fuel types would be unreliable.

Chart 3: Distribution of Selling Price (Histogram with KDE)



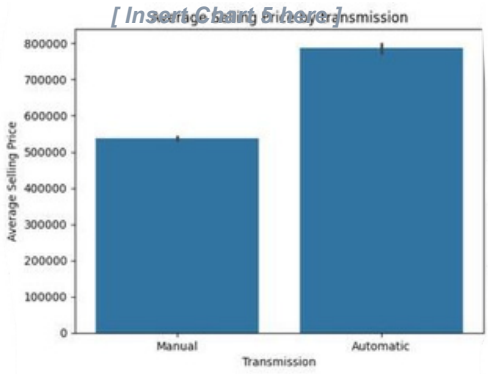
Observation: The distribution is right-skewed. Most cars sell between about 2 and 8 lakh, with the peak near 3 to 6 lakh, and the number of cars falls steadily towards the 14 lakh cut-off created by the outlier rule.

Chart 4: Max Power vs Selling Price (Scatter Plot)



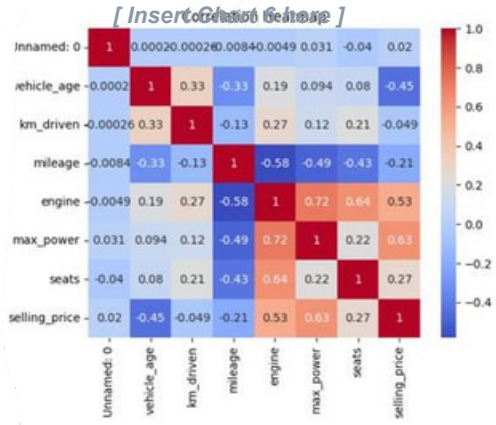
Observation: Higher max_power is generally linked with a higher selling price, but the spread is wide at every power level. The vertical stripes appear because many listings of the same model share an identical power value. A handful of cars above 300 bhp sell at modest prices and look like unusual records.

Chart 5: Average Selling Price by Transmission Type (Bar Plot)



Observation: Automatic cars have a clearly higher average selling price (roughly 7.9 lakh) than manual cars (roughly 5.4 lakh). The small error bars show the averages are estimated precisely. This shows an association only; automatics may also differ in age, power or brand.

Chart 6: Correlation Heatmap of Numerical Features



Observation:

Selling price is most positively correlated with max_power (0.63) and engine (0.53), and most negatively with vehicle_age (-0.45). km_driven has almost no linear relationship with price (-0.05). Engine and max_power are strongly related (0.72), and mileage falls as engine size grows (-0.58). The 'Unnamed: 0' row index shows no meaningful correlation.

- ***The market is dominated by petrol and diesel cars.*** Together they form about 97.7% of 15,411 listings; CNG, LPG and Electric are very rare (Chart 2).
- ***A few brands account for most listings.*** Maruti (4,992), Hyundai (2,982) and Honda (1,485) make up about 61% of the data.
- ***Selling price is right-skewed with extreme high-end outliers.*** The mean (about 7.75 lakh) is well above the median (5.56 lakh); about 9% of listings were flagged as price outliers (Charts 1 and 3).
- ***Engine power is the strongest numerical link to price.*** max_power (r = 0.63) and engine size (r = 0.53) show the highest positive correlation with selling price (Charts 4 and 6).
- ***Older cars tend to sell for less.*** vehicle_age has a moderate negative correlation with price (r = -0.45), the strongest negative link with price among the numerical features (Chart 6).

Kilometres driven alone is a weak predictor of price here. The correlation with price is only -0.05, although km_driven rises with vehicle_age (r = 0.33), so age carries more of the price information in this dataset.

Automatic cars are priced higher on average than manual cars. The average is roughly 7.9 lakh versus 5.4 lakh, while automatics are only about 21% of listings (Chart 5).

More powerful engines are less fuel-efficient. mileage is negatively correlated with engine (-0.58) and max_power (-0.49) (Chart 6).

Limitations of the dataset and analysis

1. The data is a single snapshot with no listing date, location or condition information so trends over time cannot be studied.
2. Only 4 electric, 44 LPG and 301 CNG cars are present, so nothing reliable can be said about these fuel types. Outliers were removed using the selling_price column only, which excludes premium and luxury cars from the later charts and statistics. Findings therefore describe mainstream used cars, not the whole market. Correlations measure linear association only and do not prove cause and effect. Price differences by transmission, for example, may also reflect differences in age, power or brand.
3. Units (INR, km/l, cc, bhp) are assumed because no data dictionary came with the file. Brand-wise average price and seller type were not analysed.

Conclusion

Overall learning. The CarDekho dataset was already clean: no missing values, no duplicates and consistent labels. The main data-quality issue was extreme high-end selling prices, handled with the IQR rule, which reduced the data from 15,411 to 14,025 rows. The analysis shows that listed price is mainly linked to engine power, vehicle age and transmission type, while the mix of listings is dominated by petrol and diesel cars from a few brands.

Challenges faced. Deciding how to treat outliers without throwing away useful information, reading price relationships carefully without claiming causation, and keeping the conclusions inside what the columns can actually support.

Possible future analysis. Compare average price by brand and model, apply a log transformation to price instead of removing outliers, study price by seller type and by fuel type within each brand, examine how age and km_driven interact, and build a regression model to quantify the effect of each feature.